

Classification of Eavesdropper Scenarios in Anti-Drone Physical Layer Security

Your Name

Department of Electronics and Communication Engineering

Your College Name

Your City, India

Email: your.email@example.com

Abstract—In anti-drone communication systems, the identification of an eavesdropper is a critical requirement for improving physical layer security. An eavesdropper may appear in different forms depending on its mobility, activity, identity behavior, number of attacking nodes, and operating condition. This document presents a structured classification of possible eavesdropper scenarios, including static, dynamic, passive, active, impersonating, jamming, multiple, and intermittent eavesdroppers. The classification helps in selecting suitable identification methods such as received signal strength analysis, channel state information monitoring, angle of arrival estimation, RF fingerprinting, secrecy capacity evaluation, and authentication-based verification.

Index Terms—Anti-drone system, eavesdropper identification, physical layer security, UAV communication, channel-based authentication, RF fingerprinting.

I. INTRODUCTION

In an anti-drone communication system, the legitimate transmitter is generally represented as Alice, the legitimate receiver as Bob, and the unauthorized receiver or attacker as Eve. The main challenge is that Eve may not always behave in a fixed or simple manner. In some cases, Eve may only listen silently, while in other cases it may actively transmit, jam the link, impersonate Bob, or cooperate with other eavesdroppers.

Therefore, before designing an Eve identification model, it is important to classify all possible Eve scenarios. Such classification provides a clear understanding of the attacker behavior and helps in selecting proper detection and authentication techniques.

II. DISCUSSION

From Table I, it can be observed that Eve identification should not depend on a single parameter. For example, received signal strength alone may not be sufficient because a moving drone, a hovering drone, and a fake Bob may produce similar power levels under different conditions. Therefore, a reliable identification system should combine multiple parameters such as channel state information, received signal strength, angle of arrival, Doppler shift, mobility pattern, RF fingerprint, bit error rate, signal-to-noise ratio, secrecy capacity, and authentication response.

The most difficult case is passive Eve because it only listens and does not transmit any detectable signal. In contrast, active Eve, jamming Eve, and impersonating Eve are relatively easier

to identify because they create abnormal signal patterns, timing mismatch, or authentication failure.

III. CONCLUSION

This document presented a professional classification of possible eavesdropper scenarios in an anti-drone physical layer security system. Eve may be static, dynamic, passive, active, jamming, impersonating, relaying, single, multiple, or intermittent. In the special case where only one drone is present, the system must verify whether it is Bob or Eve using identity, channel, location, mobility, and RF fingerprint information. Hence, Eve identification should be performed using a combined decision approach instead of depending on only one detection parameter.

REFERENCES

- [1] A. D. Wyner, "The wire-tap channel," *Bell System Technical Journal*, vol. 54, no. 8, pp. 1355–1387, 1975.
- [2] M. Bloch and J. Barros, *Physical-Layer Security: From Information Theory to Security Engineering*. Cambridge University Press, 2011.
- [3] Y. Zeng, J. Lyu, and R. Zhang, "Cellular-connected UAV: Potential, challenges, and promising technologies," *IEEE Wireless Communications*, vol. 26, no. 1, pp. 120–127, 2019.
- [4] K. Merchant, S. Revay, G. Stantchev, and B. Noursain, "Deep learning for RF device fingerprinting in cognitive communication networks," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 160–167, 2018.

TABLE I: Classification of Possible Eavesdropper Scenarios in Anti-Drone Physical Layer Security

Category	Eve Scenario	Description	Possible Identification Method
Mobility-Based	Static Eve	Eve remains fixed at a particular position and observes the communication link between Alice and Bob. This type of Eve has almost constant distance and channel behavior.	RSSI analysis, CSI monitoring, location verification, secrecy capacity observation.
	Hovering Eve	Eve behaves like a drone that stays nearly fixed in air. Although it appears stationary, small hovering movements can still create minor channel variations.	Small-scale channel variation analysis, Doppler shift observation, angle of arrival estimation.
	Dynamic Eve	Eve moves continuously with time. Its position, received power, Doppler profile, and channel gain change according to its movement.	Mobility tracking, Doppler analysis, time-varying CSI, trajectory verification.
Activity-Based	Passive Eve	Eve only listens to the confidential signal and does not transmit anything. This is one of the most difficult cases because Eve may not leave any direct signal evidence.	Indirect detection using secrecy capacity, outage probability, BER, SNR, and suspicious channel behavior.
	Active Eve	Eve transmits signals into the network. It may send fake pilots, false responses, or unauthorized access requests.	Pilot authentication, RF fingerprinting, CSI mismatch detection, timing analysis.
	Jamming Eve	Eve intentionally transmits interference to reduce the received signal quality at Bob and disturb the legitimate communication link.	Sudden SNR degradation, BER increase, interference power detection, spectrum monitoring.
	Transmitting and Receiving Eve	Eve listens to Alice and also transmits signals toward Alice or Bob. It may disturb, mislead, or try to enter the communication link.	Bidirectional signal monitoring, abnormal packet detection, RF fingerprinting, traffic behavior analysis.
Identity-Based	Eve Acting as Bob	Eve tries to impersonate Bob and convince Alice that it is the legitimate receiver. This case is highly critical in command and control communication.	Channel-based authentication, location verification, RF fingerprinting, challenge-response verification.
	Relay Eve	Eve acts as an intermediate node by receiving the signal from Alice and forwarding or modifying it toward Bob. This resembles a man-in-the-middle behavior.	Latency analysis, relay path detection, timing mismatch, channel inconsistency checking.
	Unknown Drone	A detected drone has no confirmed identity. The system must decide whether it is a legitimate drone or an unauthorized drone.	Registered ID checking, channel profile matching, location confirmation, authentication verification.
	Compromised Bob	A legitimate receiver is captured, misused, or controlled by an attacker and starts behaving abnormally.	Behavior monitoring, abnormal request detection, trust score evaluation, historical profile comparison.
Number-Based	Single Eve	Only one unauthorized drone or receiver is present and tries to listen to or attack the communication link.	Single-node channel analysis, secrecy capacity estimation, received power comparison.
	Multiple Independent Eves	More than one Eve is present, but each attacker operates independently without sharing information with others.	Multi-node localization, angle of arrival estimation, distributed signal monitoring.
	Colluding Eves	Multiple Eves cooperate and share their received information to improve the chance of decoding Alice's confidential message.	Cooperative attack detection, correlation analysis, multi-receiver secrecy analysis.
Special Cases	Only One Drone Present	If only one drone is detected, the system cannot directly assume that it is Bob. It must verify whether the drone is legitimate or suspicious.	Authentication, expected trajectory checking, RF fingerprint matching, CSI comparison.
	Intermittent Eve	Eve appears only during selected time intervals and remains silent at other times to avoid continuous detection.	Time-window analysis, anomaly detection, periodic channel monitoring.